NOTRE DAME UNIVERSITY

BANGLADESH

EEE Lab Report-07

Course Title: Electrical Circuit Analysis Lab

Course Code: EEE-2108

Experiment No: 07

Name of the Experiment: Determining the I-V characteristics of a Zener Diode & calculate the line and load Regulation.

Date of Experiment: 24-08-23

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Submitted by:

Group: E

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Batch: CSE-20

Submitted To:

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Lecturer, NDUB

Experiment No 07

Name of Experiment: Determining the I-V characteristics of a Zener Diode & calculate the line & load regulation.

Theory: The diodes we have studied before do not operate in the breakdown region because this may damage them. A zener diode is different; it is a silicon diode that the manufacturer has optimized for operation in the breakdown region. It is used to build voltage regulator circuits that circuits that hold the load voltage almost constant despite large change in line voltage and load resistance. The symbol of zener diode shows in figure ~~7.1~~ 7.1

Figure: 7.1 Symbol of Zener Diode.

The zener diode may have a breakdown voltage from about 2 to 200 volts. These diodes can operate in any of three regions, - forword, leakage & breakdown. Figure 7.5 shows the I-V characteristics curve of zener diode.

- In the forward region it works as an ordinary diode.
- In the leakage region it has only a small reverse saturation current.
- In the breakdown it has a sharp knee, followed by an almost vertical increase in current without changing the voltage.
- The voltage is almost constant, approximately equal to V_Z over most of the breakdown region

Equipments & Components:

1. Zener diode - 1piece
2. Resistor - 470Ω - 1 piece
3. Trainer board
4. DC power supply
5. Digital Multimeter
6. Chords & wire as required
7. Digital Storage Oscilloscope

Experimental Setup:

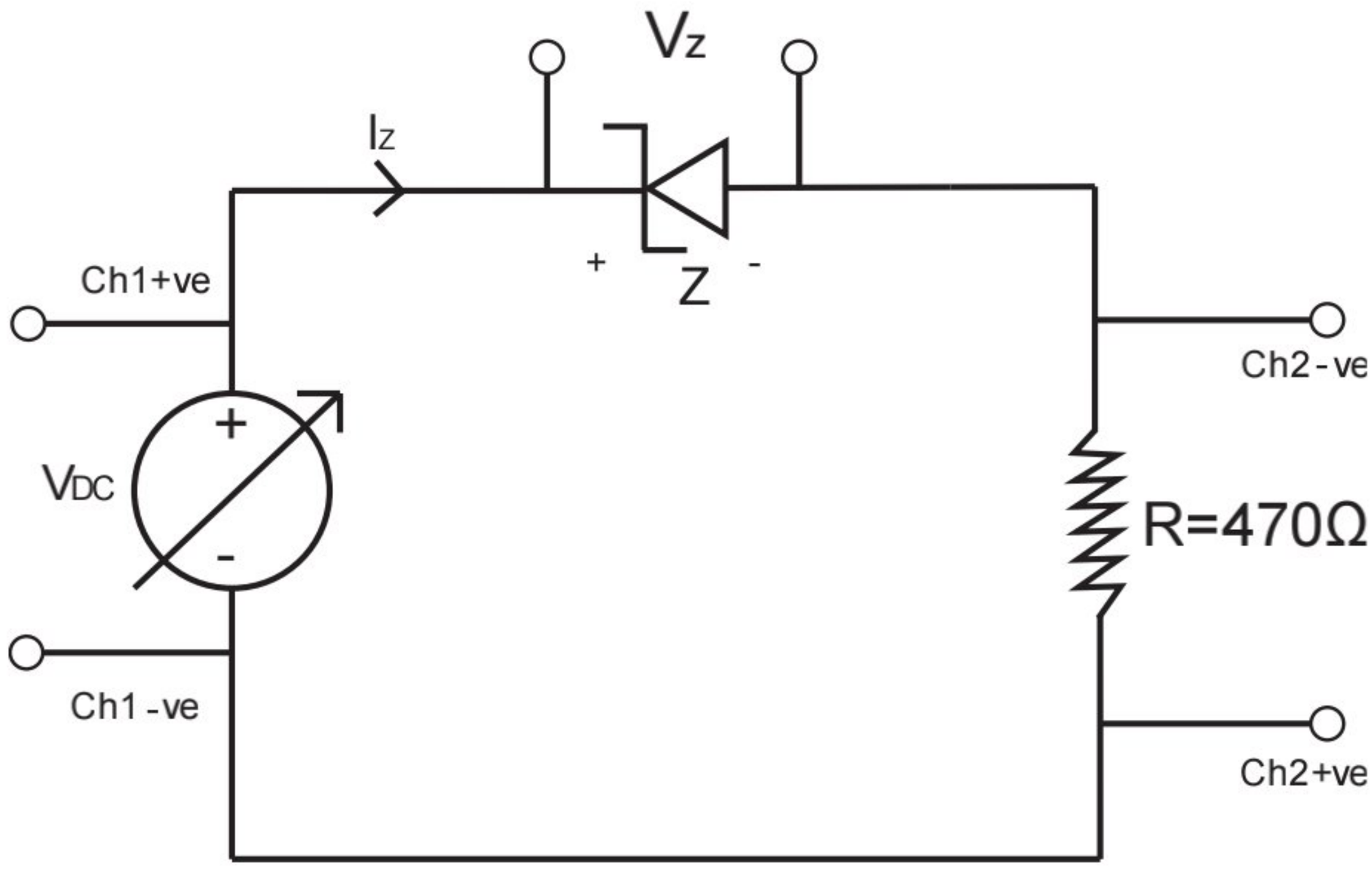


Fig 7.2: Experimental Zener Diode Circuit Design

Procedures:

1. Connect the circuit as shown in the figure 7.2.
2. Vary the supply voltage from zero volt, complete the Table.
3. Connect the circuit as shown in the figure.
4. Set the DC voltage source and change it to 0 Volt to 15 volt.
5. Varying the supply voltage, for individual voltage measure the current and fill in table.
6. Calculate the current of zener diode using the formula $I_d = \frac{V_{DD} - V_Z}{R}$ and fill in table.
7. Now connect the circuit according to Fig 7.2. Set the Digital Storage Oscilloscope in X-Y mode. We will be able to see the I-V characteristic curve in the DSO.

Experiment No: 07

Group: E

Data sheet:

ID: 5, 2, 9, 14

	V_{DD} (v)	V_Z (v)	Measured current I_Z (mA)	Calculated current $I_d = \frac{V_{DD} - V_Z}{470\Omega}$
1.	0	-0.004	0	8.51×10^{-3}
2.	0.5	0.389	0	0.23617
3.	1.0	0.934	0	0.14042
4.	1.5	1.49	0	0.2127
5.	2	1.94	0	0.00012765
6.	2.5	2.47	0	0.00006382
7.	3	2.904	0	0.00020425
8.	3.5.	3.435	0	0.00013829
9.	4	4.03	0	0.00006382
10.	4.5	4.48	0	0.00042553
11.	5	4.93	0.03 mA	0.0094894
12.	5.4	5.14	0.55 mA	0.00055319
13.	5.8	5.16	1.43 mA	0.00136170
14.	6.2	5.17	2.30	0.00219148
15.	6.6	5.17	3.13	0.00304255
16.	7	5.17	3.93	0.00389362
17.	7.4	5.18	4.84	0.00472340
18.	7.8	5.18	5.74	0.00557447
19.	8.2	5.18	6.67	0.00642553
20.	8.6	5.19	7.53	0.00725532
21.	9	5.19	8.30	0.00810638
22.	9.4	5.19	9.32	0.00895745
23.	9.8	5.20	10.24	0.009787234
24.	10.2	5.20	11.08	0.010638298
25.	10.6	5.20	11.85	0.011489362
26.	11	5.21	12.72	0.012319149
27.	11.4	5.21	13.63	0.013170213
28.	11.8	5.21	14.49	0.014021277
29.	12.2	5.21	15.40	0.014872234
30.	12.6	5.22	16.36	0.015702128
31.	13	5.22	17.17	0.019655319
32.	13.4	5.22	17.99	0.017404255

33.	13·8	5·23	19·05	
34.	14·2	5·23	19·74	
35.	14·6	5·23	20·70	
36.	15	5·23	21·56	

29/08/23

Load Regulation $= \frac{.09\ V}{21\ mA}$

$= \frac{.09\ V}{21 \times 10^{-3}\ A}$

$= \frac{90}{21}\ \frac{V}{A}$

Line reg. $= \frac{\Delta V_{out}}{\Delta V}$

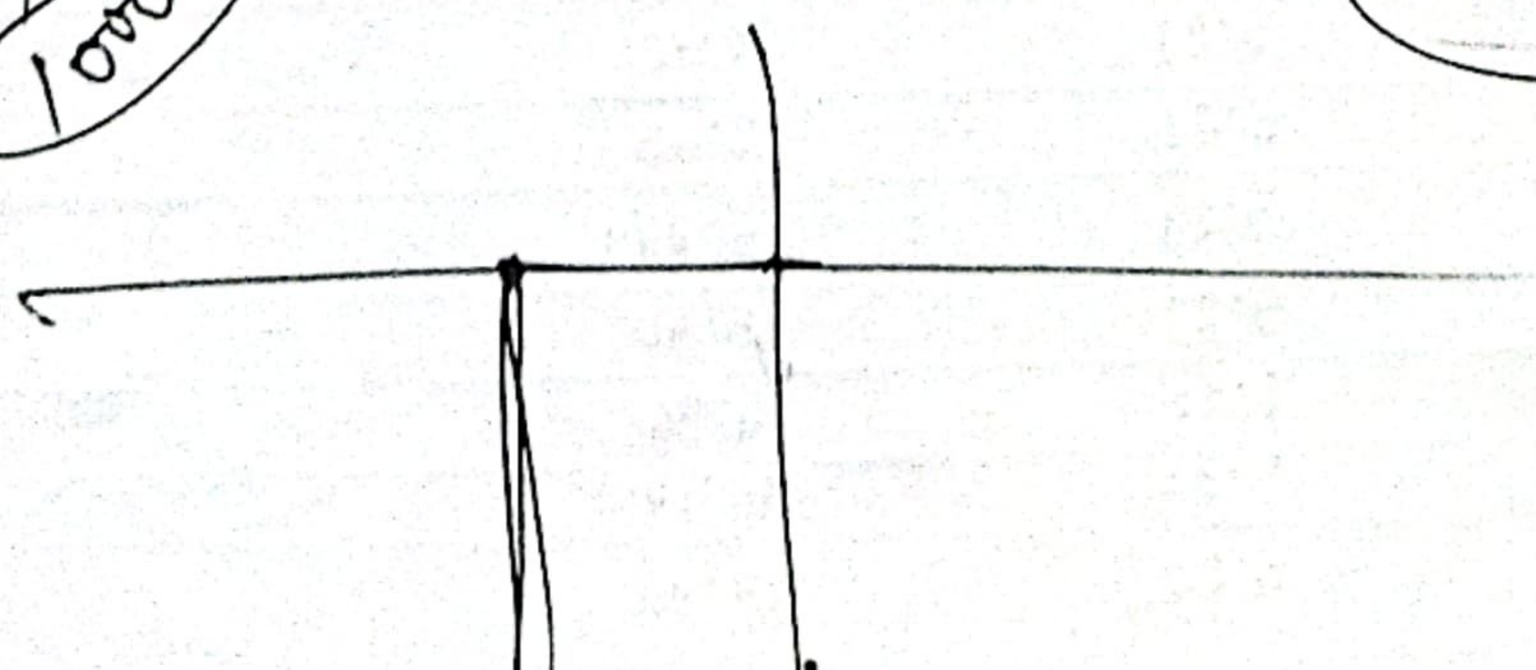
$= \frac{.09}{10}$

$= \frac{9}{1000}$

$.009$

$V_2 = V_{20} + I_2 r_2$

$= 4 \cdots \Omega$



Student ID: 2, 5, 9, 14

Group No: E

DATA SHEET:

Table: Measuring current through the Diode (R=470Ω)

Serial no.	DC Supply Voltage (V_{DD}) (V)	Measured Voltage of Zener Diode (V_z) (V)	Measured Current of Zener Diode (I_z) (mA)	Calculated Current of Zener Diode $I_{z_c} = \frac{V_{DD}-V_z}{R}$ (mA)
1.	0	-0.004	0	0.00000851
2.	0.5	0.389	0	0.00023617
3.	1	0.934	0	0.00014042
4.	1.5	1.49	0	0.00002127
5.	2	1.94	0	0.00012765
6.	2.5	2.47	0	0.00006382
7.	3	2.904	0	0.00020425
8.	3.5	3.435	0	0.00013829
9.	4	4.03	0	-0.00006382
10.	4.5	4.48	0	0.00042553
11.	5	4.93	0.03	0.00014894
12.	5.4	5.14	0.55	0.00055319
13.	5.8	5.16	1.43	0.00136170
14.	6.2	5.17	2.30	0.00219148
15.	6.6	5.17	3.13	0.00304255
16.	7	5.17	3.93	0.00389362
17.	7.4	5.18	4.84	0.00472340
18.	7.8	5.18	5.74	0.00557447
19.	8.2	5.18	6.67	0.00642553
20.	8.6	5.19	7.53	0.00725532
21.	9	5.19	8.30	0.00810638
22.	9.4	5.19	9.32	0.00895745
23.	9.8	5.20	10.24	0.009787234
24.	10.2	5.20	11.08	0.010638298
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28.	11.8	5.21	14.49	0.014021277
29.	12.2	5.21	15.40	0.01487234
30.	12.6	5.22	16.36	0.015702128
31.	13	5.22	17.17	0.016553191
32.	13.4	5.22	17.99	0.017404255
33.	13.8	5.23	19.05	0.018234043
34.	14.2	5.23	19.74	0.019085106
35.	14.6	5.23	20.70	0.01993617
36.	15	5.23	21.56	0.020787234

Expected Waveforms:

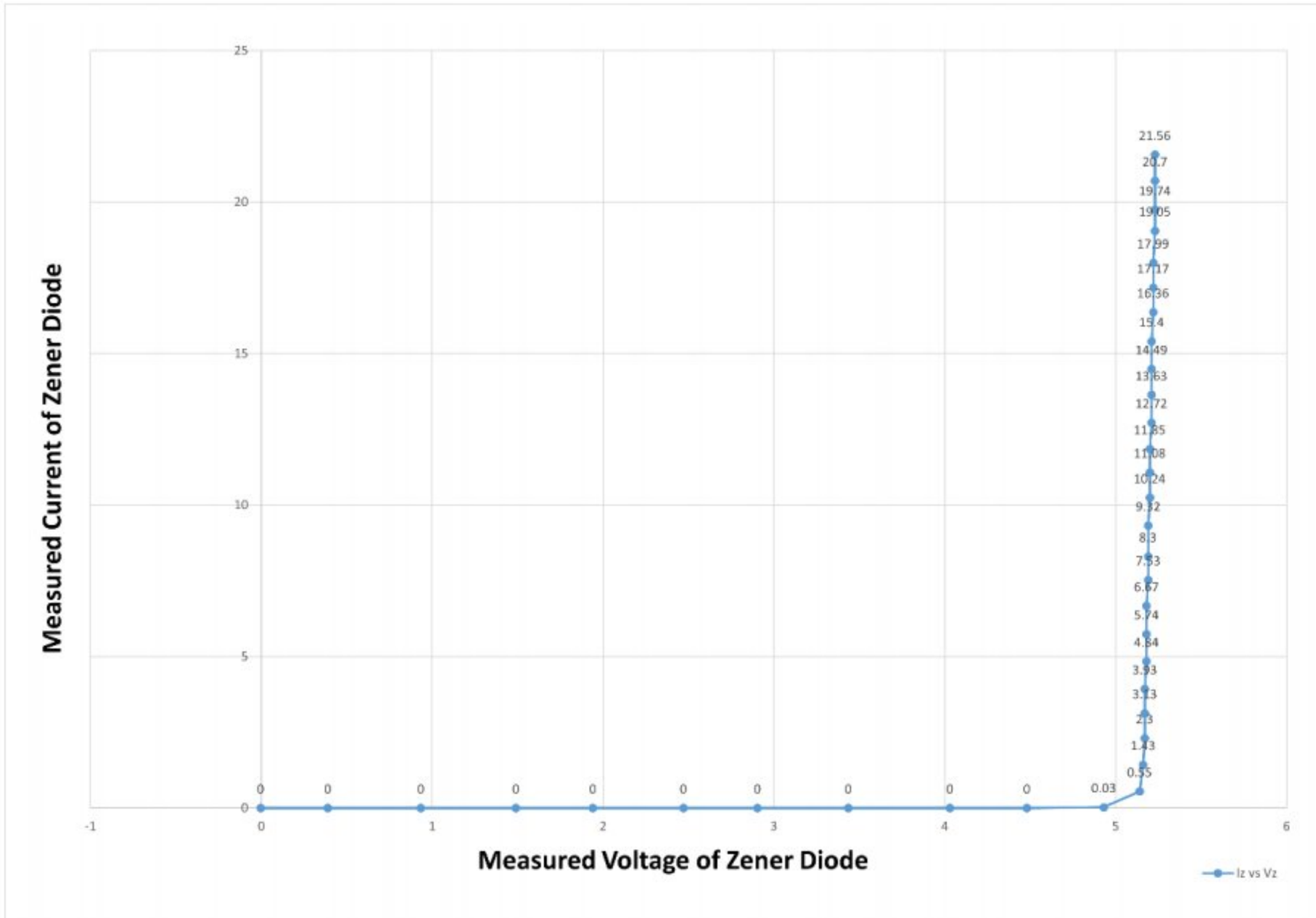

Fig 7.3: I_z vs V_z Graph

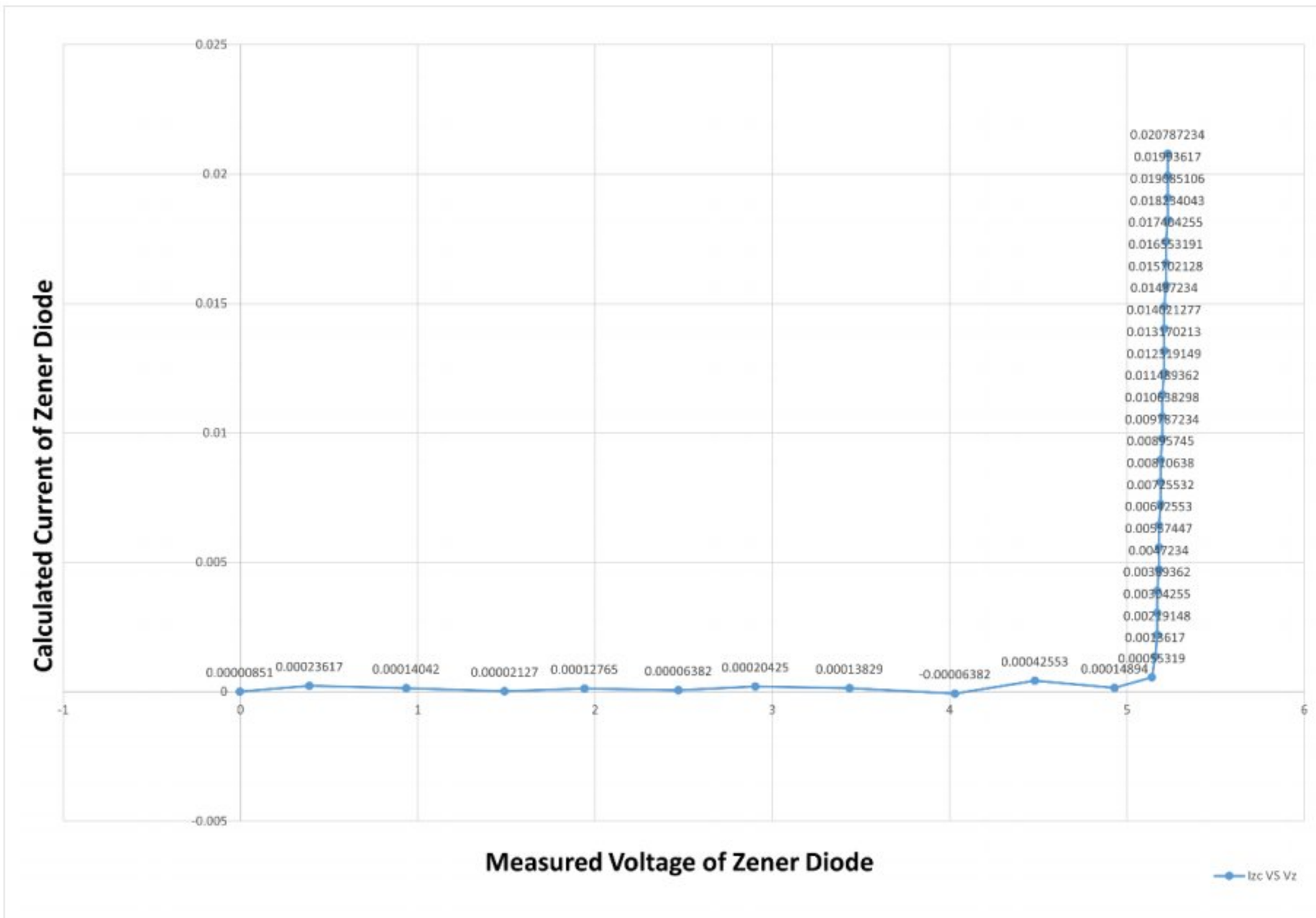

Fig 7.4: I_{z_c} vs V_z Graph

I-V Characteristics Curve:

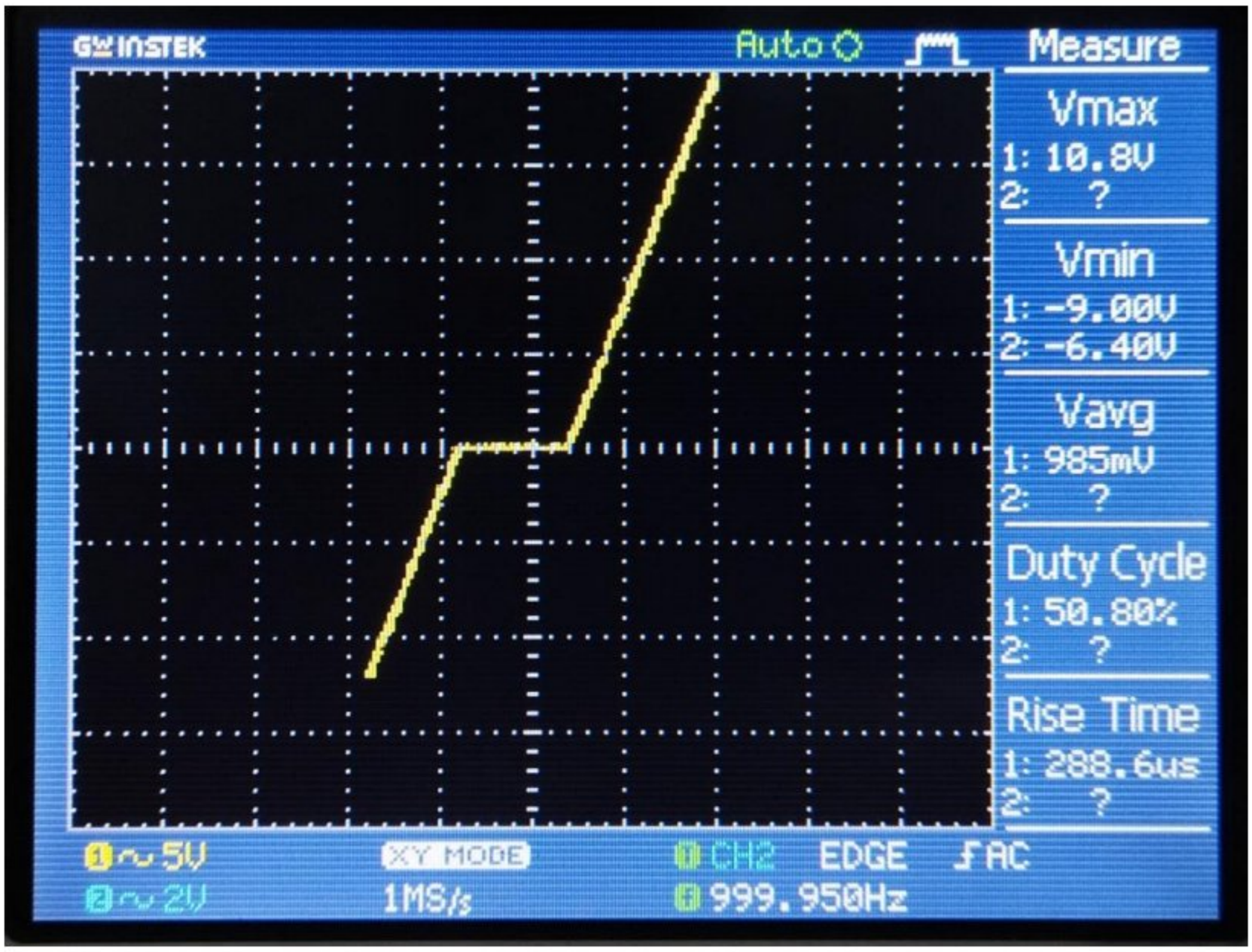

Fig 7.5: I-V Characteristic Curve in X-Y Mode

Discussion :

1. Calibrate the oscilloscope for accuracy.
2. Use low voltage & current.
3. Be contious with reverse voltage to prevent damage.
4. Measure voltage in parallel & when measuring current, short the circuit.
5. Avoid overloading & short circuit.
6. Follow guidelines & manual.